

WEEKLY SCHEDULE SPR/SUM 2026

(also known as Course Section Information)

CST8917 Serverless Applications **Section 300 – Hybrid** **Cloud Development and Operations – Level 2**

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Learning Resources

Required Textbook:

- All required texts and resources will be provided in Brightspace.

Recommended Resources:

- All recommended texts and resources will be provided in Brightspace.

Hardware and Software:

- Cloud Service Provider (CSP) subscription (Free Tier)
- A computer machine (physical or virtual) with the following minimum system requirements:
 - OS: Ubuntu 22.04.1 LTS
 - 2 GHz dual-core processor or better
 - 4 GB system memory or better
 - 80 GB of free hard drive space or better

Evaluation Breakdown

Assessment	Due Date and Time	Value	CLRs
4 Quizzes (at 7.5% each)	Week 4, 6, 10, 12	30%	1,2,3
4 Labs (at 7.5% each)	Week 3, 5, 9, 11	30%	1,2,4
2 Assignments	Week 4, 12	15%	1,2,3,4
Midterm Project	Week 7	15%	1,2,3,4
Final Oral Exam	Week 14, Week 15	10%	1,2,4

Learning Schedule

LEARNING SCHEDULE IS SUBJECT TO CHANGE WITH NOTIFICATION

Date	Weekly Theme and Learning Outcomes	Learning Activities	Assessments (%) Due Date	CLRs
Week 1 May 15	Course Introduction - Understand course scope - Serverless Computing	Provided in Brightspace.		1, 2
Week 2 May 22	Azure Functions - Triggers: HTTP, Timer, Blob - Bindings: Input/Output	Provided in Brightspace.		1,4
Week 3 May 29	Azure Function Chaining & Orchestration - Fan-out/fan-in - Durable Functions	Provided in Brightspace.	Lab 1 (7.5%)	1,4
Week 4 June 5	Long running Azure Functions - Long-running Workflows - Durable Functions Deep Dive	Provided in Brightspace.	- Quiz 1 (7.5%) - Assignment 1 (5%)	1,4
Week 5 June 12	Development, Deployment, & Testing I - VS Code integration - Testing & local debugging, CI/CD	Provided in Brightspace.	Lab 2 (7.5%)	1,2,4
Week 6 June 19	Development, Deployment, & Testing II - VS Code integration - Testing & local debugging, CI/CD	Provided in Brightspace. Discuss Mid-term Project		1,2,4
Week 7 June 26	Midterm Project (Group Assignment)		Midterm Project (15%)	
Week 8 July 3	BREAK	BREAK	BREAK	
Week 9 July 10	Logic Apps & Workflow Automation - Logic Apps vs Durable Functions - Triggers, connectors, and real-world cases	Provided in Brightspace.		2,4
Week 10 July 17	Messaging Services – Service Bus & Storage Queues - Durable messaging - Queue-triggered functions	Provided in Brightspace.	Quiz 2 (7.5%)	2,4
Week 11 July 24	Event Ingestion and Routing Event Hubs, Event Grid, Azure Relay	Provided in Brightspace.	Lab 3 (7.5%)	2,4
Week 12 July 31	Use Case Deep Dive Real-world app scenario	Provided in Brightspace.	Quiz 3 (7.5%)	2,4
Week 13 August 7	Architectural Patterns & Best Practices When to use Functions vs Logic Apps vs Durable	Provided in Brightspace.	Lab 4 (7.5%)	1,2,3, 4
Week 14 August 14	Final Oral Exam (10%)	Provided in Brightspace.	Final Assignment 2 (10%)	2,4

Pass condition

In order to pass the course, students must:

- Achieve at least **50%** on the total final grade of the course.
- Meet all the following minimum thresholds for key assessment categories:
 - **Quizzes:** A minimum average of **50%**.
 - **Labs:** A minimum average of **50%**.
 - **Final Oral Exam:** A minimum of **50%**.

Achieving a higher mark in one evaluation area **can not be used** to offset a low mark in the other. All students are required to write quizzes, midterm and final exams. **There are no provisions** for “making up” a missed quiz or exam. If, as a result of being off-track in your program or some unforeseen circumstance, you note that there is a scheduling conflict in your final exam schedule, it is your responsibility to alert your course professor no later than one week before final exams start, to allow for any special arrangements.

Other Important Information

Attendance to Class: Students are required to attend class and online session each week. The nature of the course content requires students to engage in class discussions, group work and collaborative activities. In-class completion of evaluated tasks and short assignments maybe required. Students who regularly attend classes are more likely to be successful. When students are unable to attend, they are responsible for obtaining any missed information.

Late Assignments: Assignments are due on the date and time specified by the course professor. Assignments not submitted on the assigned date and time will receive a deduction of 10% per day. Students will receive a zero after 5 days. All assignments should be prepared as per the direction of the professor. Assignments should be clearly and neatly written, show all work, and reflect College standard and a professional image.

Academic Integrity: As a learner you are expected to adhere to Algonquin's Academic Integrity Policy AA48 while completing your course work. If academic misconduct is suspected, the incident will be reported to the Academic Integrity Manager, and you may receive a zero on the assignment or potentially a more severe academic sanctions.

Classroom Technology Use: Laptops will only be used when they are needed for an in-class exercise. Cell phones must be silenced and put away at the beginning of class, unless you are expecting an emergency call.

Course Policy on Generative AI Use: Use Permitted

The educational value of this course lies not only in the mastery of course content but also in the process of engaging critically and creatively with the material. Generative AI tools can be powerful aids in the learning process when used responsibly and ethically.

Permitted Use:

- You are allowed to use generative AI tools (e.g., ChatGPT, GitHub Copilot, etc.) to assist with brainstorming, drafting, coding, or problem-solving, provided that their use is fully disclosed and appropriately cited.
- AI tools may be used to support your understanding of the course content, generate ideas, or help troubleshoot code. However, these tools should not replace your own critical thinking, problem-solving, or the creative processes that are integral to your learning.

Assessments Where AI Use is Permitted:

AI tools are only allowed in the following assessments:

- **Assignments**
- **Labs**
- **Midterm Project**

Students must disclose and appropriately cite their use of AI in these assessments, specifying how the tools were used and ensuring that the final submission reflects their own understanding and effort.

Prohibited Use

Generative AI tools are strictly prohibited in the following assessments:

- **Quizzes**
- **Final Oral Exam**

Using AI tools in these assessments will be considered a violation of the course policy and academic integrity guidelines.

Requirements:

- **Transparency:** If you use generative AI in any of your submissions, you must disclose this in your work. This includes specifying how the AI was used and including citations where appropriate. For example, if you used AI to draft parts of an essay, note this in a footnote or endnote.
- **Original Thought:** The work you submit must primarily reflect your own understanding and effort. AI-generated content should not be submitted as your own without significant modification and integration into your work.
- **Critical Evaluation:** Any content produced or suggested by AI should be critically evaluated and not accepted at face value. You are responsible for ensuring the accuracy and relevance of all material included in your submissions.

Academic Integrity:

- **Prohibited Use:** Directly submitting AI-generated work as your own without modification, evaluation, or proper citation is considered a violation of academic integrity under Algonquin College Policy AA48. This is akin to contract cheating, where work produced by a third party is submitted as your own.
- **Consequences:** Violations of this policy will be treated as instances of academic misconduct. This could result in receiving a zero on the assignment, further academic penalties, or other disciplinary actions as outlined in the college's academic integrity policy.

By responsibly incorporating generative AI into your learning process, you can enhance your educational experience while maintaining the integrity and value of your work.